

**AI-Powered Mobile Platform Enhancing Public Safety and
Community Policing in Sri Lanka**

2526-J200

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology specializing in
Information Technology.

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

July 2025

Real-Time AI System for Detecting Harmful Social Media Content
2526-J200

Project Proposal Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

The rise of social media has significantly increased the spread of harmful content, including harassment, killings, elder abuse, child abuse, and violent incidents. Manual moderation methods are inefficient given the volume and speed of online content, while existing single-modality AI approaches struggle to detect complex harmful behaviors. This research introduces a real-time multimodal artificial intelligence framework that analyzes text, video, and voice from social media posts to identify harmful activities. The system integrates natural language processing (NLP) for Sinhala and English text and audio transcript analysis, computer vision models for violent scene and action recognition, and speech processing for semantic and sentiment cues. A two-stage pipeline is proposed keyword and hashtag-based pre-filtering in Sinhala and English to select potentially harmful posts, and deep learning-based multimodal fusion models to classify harmful content with high accuracy and reduced false positives. Designed for low-latency, real-time detection, the framework automatically flags harmful content and issues alerts to moderators or law enforcement. The novelty lies in its bilingual multimodal approach, continuous adaptation to evolving slang and abusive language in Sinhala and English, and the fusion of text, audio, and video streams to ensure robust detection in real-world social media environments. This research highlights the potential of multimodal AI to enhance online safety, enable timely interventions, and support public protection in the digital space.

Keywords: Multimodal AI, Natural Language Processing (NLP), Sinhala-English bilingual analysis, Computer Vision, Speech recognition, Harmful content detection, Deep learning, Public safety

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
NLP	Natural Language Processing
CV	Computer Vision
ASR	Automatic Speech Recognition
SI	Sinhala
EN	English
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Approach
XLM-R	Cross-lingual Language Model – RoBERTa
YOLOv8	You Only Look Once (v8) real-time object/action detection
SlowFast	Two-pathway video recognition network
Video Swin	Video Swin Transformer (video recognition)
Whisper	OpenAI speech-to-text model
wav2vec 2.0	Self-supervised speech representation model
STT	Speech-to-Text
ONNX	Open Neural Network Exchange
API	Application Programming Interface
GPU	Graphics Processing Unit
UAT	User Acceptance Testing
RBAC	Role-Based Access Control

1. INTRODUCTION

Social media has become one of the most influential platforms in today's digital era, enabling millions of people to share their thoughts, experiences, and multimedia content in real time [1]. While these platforms foster communication and awareness, they have also become channels where harmful and criminal activities are exposed. In Sri Lanka, many incidents of harassment, killings, elder abuse, child abuse, and violent acts are first posted on social media rather than being reported directly to authorities. This creates a significant delay in law enforcement responses, limiting the ability of police and related agencies to intervene quickly and prevent further harm [1].

The traditional method of monitoring harmful activities relies heavily on manual inspection of social media posts, which is impractical given the sheer volume of daily uploads. Police officers and analysts cannot feasibly review millions of posts, images, and videos in real time, leading to gaps in crime detection and delayed responses. Moreover, manual inspection is prone to human error, bias, and oversight, which further reduces its effectiveness in detecting subtle yet serious cases of abuse and violence [21]. To overcome these limitations, modern technologies such as Artificial Intelligence (AI), Natural Language Processing (NLP), Computer Vision, and Speech Analysis are increasingly being explored for automated detection of harmful content. AI-driven systems can process large volumes of text, video, and audio data simultaneously and identify patterns that may indicate harassment, abuse, or violent activities; importantly, multimodal approaches improve robustness over single-modality methods [11].

This study proposes a multimodal AI-based system that can analyze text, video, and voice content in social media posts to automatically detect categories such as harassment, killings/murder acts, elder abuse, child abuse, and violent behavior. The system integrates Natural Language Processing for text analysis, Computer Vision for video content analysis, and Speech Recognition with sentiment/emotion analysis for voice data, leveraging established models in each area [2], [5], [7]. By combining these modalities, the proposed approach ensures higher accuracy and reliability in identifying harmful content.

Furthermore, the research emphasizes the importance of tailoring solutions to the Sri Lankan context, focusing on bilingual language support (Sinhala and English), which is

critical for understanding local posts and discussions [14]. The envisioned system would enable law enforcement agencies and social media platforms to act proactively, reducing delays in identifying threats, improving public safety, and ultimately saving lives [11]. Finally, this proposal highlights the necessity of adopting AI-based real-time monitoring solutions to strengthen social security in Sri Lanka. By addressing the challenges of manual monitoring and leveraging cutting-edge AI techniques, the proposed system aims to reshape the way authorities respond to online crime-related content. The outcome is expected to empower law enforcement with timely insights, support preventive measures, and contribute significantly to building a safer digital and physical environment [11].

1.1 Background & Literature survey

Social media has become one of the most dominant communication platforms worldwide, enabling users to instantly share thoughts, opinions, and multimedia content [1]. While it offers significant benefits in terms of awareness and connectivity, social media has also become a channel where harmful and crime-related content spreads rapidly. Categories such as harassment, hate speech, child abuse, elder abuse, and violent acts are frequently observed in online spaces, creating serious threats to mental health, social stability, and public safety. In the Sri Lankan context, the issue is particularly critical. Social media platforms such as Facebook, YouTube, TikTok, and Twitter/X are widely used for communication and entertainment [1]. However, many crimes are first posted on these platforms instead of being reported directly to authorities. This results in significant delays for police intervention, limiting their ability to act quickly and prevent further harm. The survey conducted among social media users confirmed this reality: 92.9% of respondents stated that they have encountered harmful content online as shown in Figure 1.1 below.

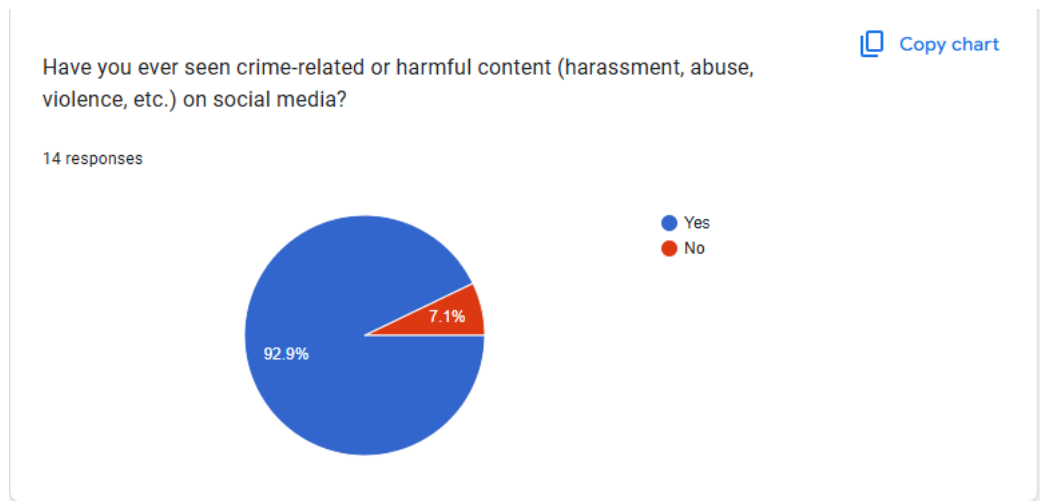


Figure 1.1: Exposure to harmful content on social media

Among these, hate speech (92.9%) and harassment/bullying (85.7%) were the most commonly observed, followed by child abuse (42.9%), violent acts/murder (42.9%), and elder abuse (35.7%) as shown in Figure 1.2 below.

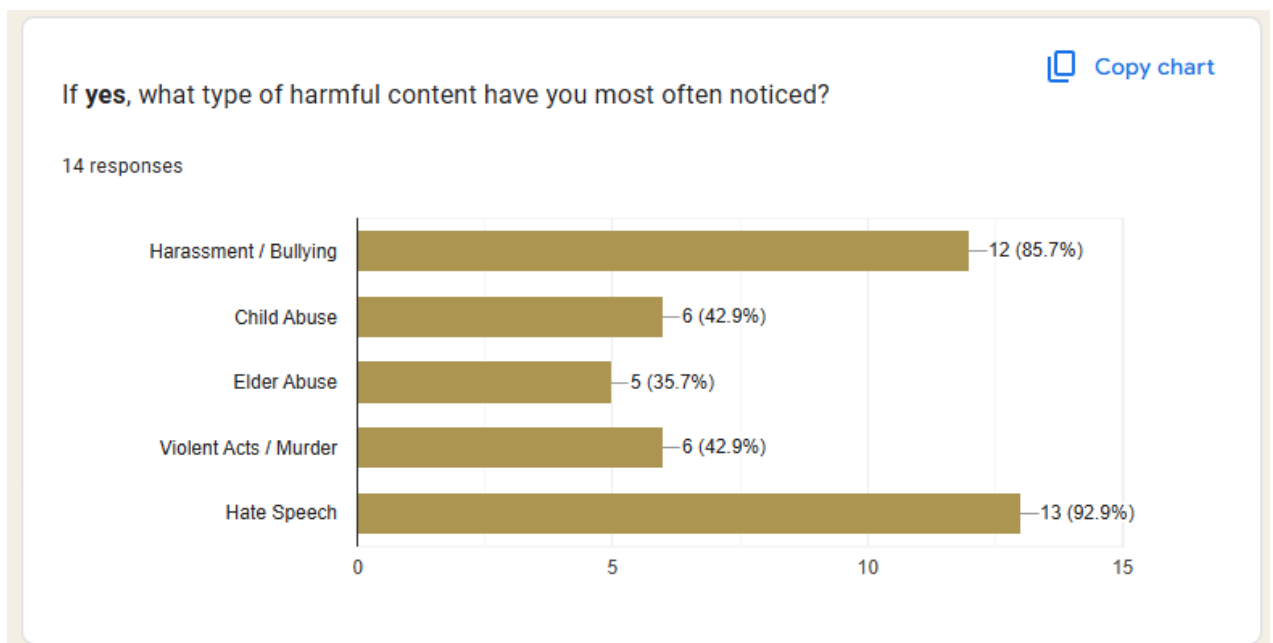


Figure 1.2: Types of harmful content most often noticed

Furthermore, 57.1% of participants reported encountering harmful content occasionally, while 28.6% reported seeing it very frequently as shown in Figure 1.3 below.

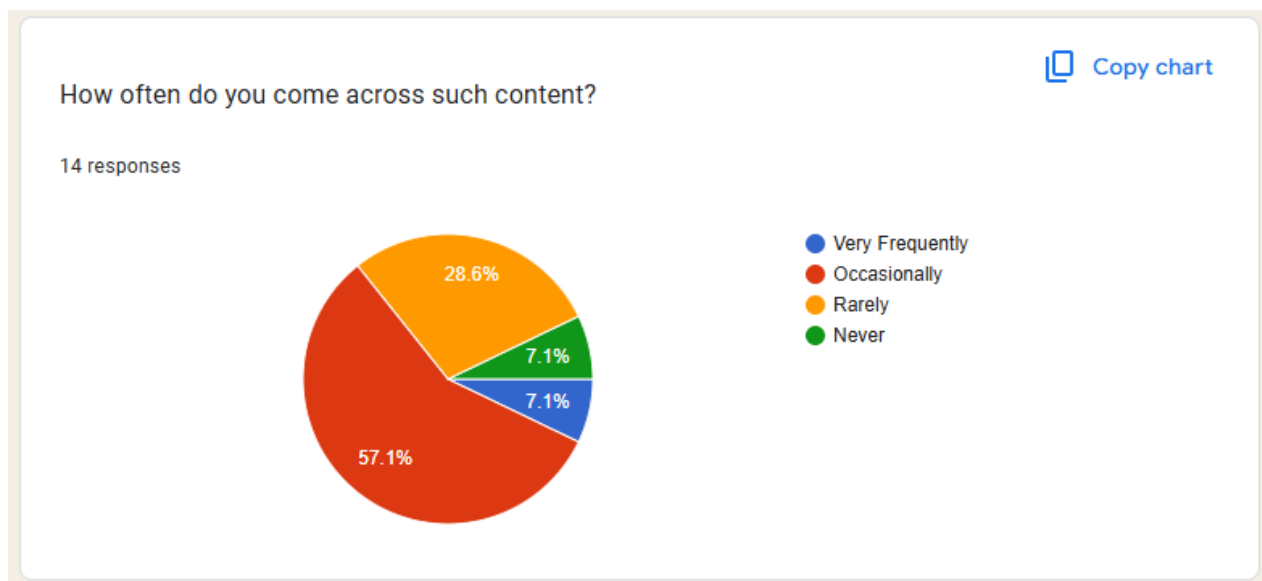


Figure 1.3: Frequency of encountering harmful content.

These findings highlight the prevalence of harmful content and validate the urgency for automated detection systems.

Despite the seriousness of the issue, existing methods of monitoring rely heavily on manual inspection or user reports. This is impractical due to the massive volume of daily uploads and is prone to human error, bias, and oversight [21]. The survey further emphasized this challenge, with 71.4% of participants stating that Sri Lankan police either rarely respond quickly or usually do not respond at all to crime-related posts as shown in Figure 1.4 below.

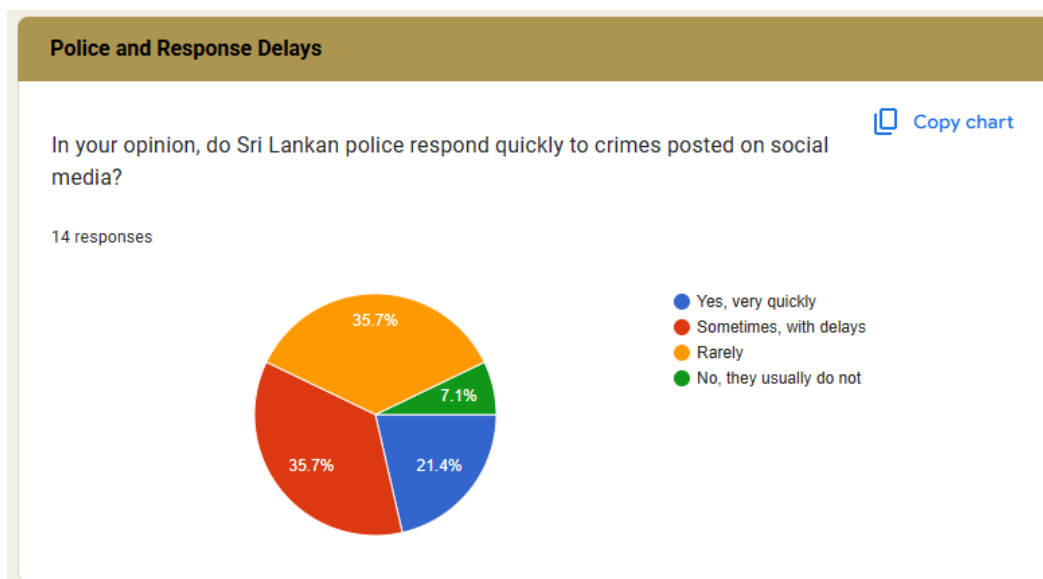


Figure 1.4: Public perception of police response to crime-related social media posts

In addition, 71.5% agreed or strongly agreed that delays in seeing crime-related posts reduce the ability of police to prevent crime effectively as shown in Figure 1.5 below.

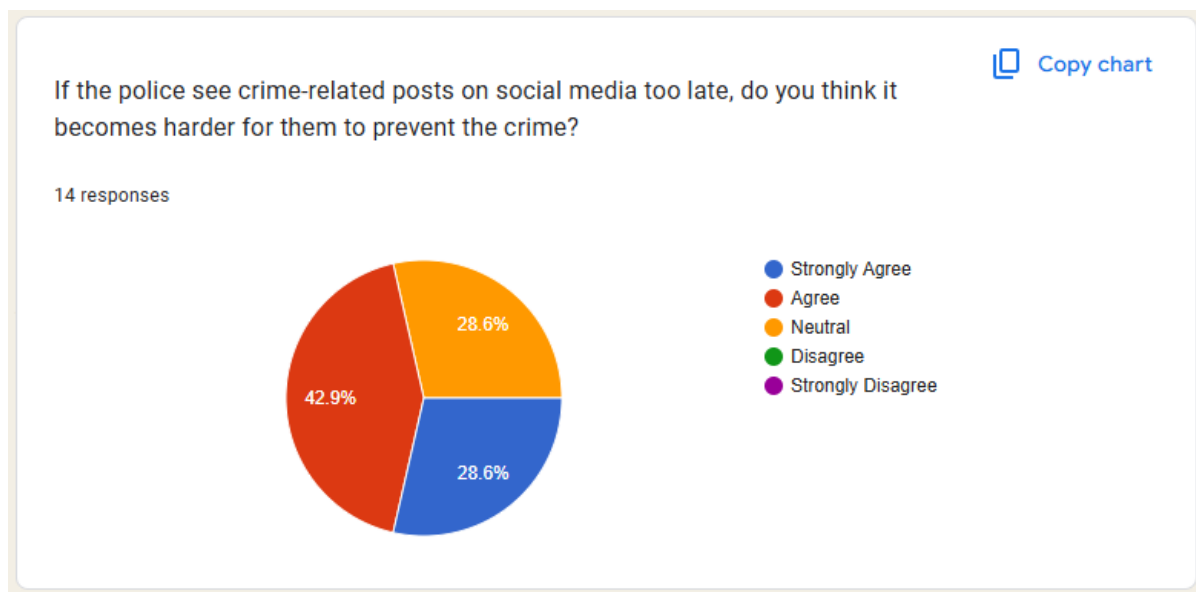


Figure 1.5: Perceived impact of delayed visibility on crime prevention

Encouragingly, 85.7% of respondents supported the implementation of an AI-based system that automatically detects and reports harmful or crime-related content on social media as shown in Figure 1.6 below.

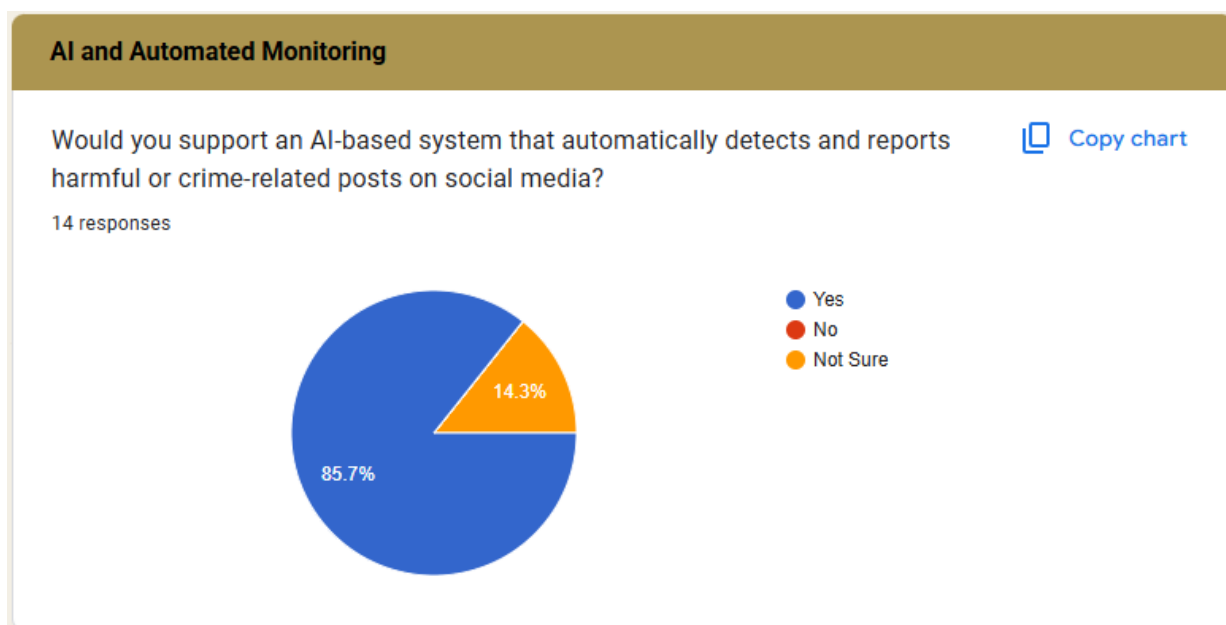


Figure 1.6: Public support for an AI-based automated harmful content detection system.

Globally, researchers have investigated artificial intelligence based approaches to detect harmful online content. Text-based approaches using Natural Language Processing (NLP)

models such as BERT, RoBERTa, and XLM-R have shown strong performance in detecting harassment, hate speech, and abusive language [2],[4]. Computer Vision (CV) models, such as YOLO, SlowFast, and Video Swin Transformer, have been used to detect violent actions in videos [7],[9], while Audio Analysis models such as Whisper and wav2vec 2.0 have been used for speech-to-text transcription and sound event detection [5], [6]. Recent work has demonstrated that multimodal approaches, which combine text, video, and audio features, achieve higher accuracy and robustness compared to single-modality systems [11], [12], [19].

However, most of these systems have been developed in Western contexts and high-resource languages, with limited application in low-resource, bilingual environments like Sri Lanka. Very few systems integrate real-time fetching of social media posts with multimodal detection across Sinhala and English, leaving a significant gap in the area of law enforcement support and automated monitoring[14].

Therefore, this research proposes a real-time multimodal AI system that fetches and analyzes social media posts in Sinhala and English, automatically detecting harassment, killings, elder abuse, child abuse, and violent activities by combining NLP, computer vision, and speech recognition techniques. This approach directly addresses the gap identified in both the global literature and the local survey findings, providing a context-aware, bilingual, and real-time solution to strengthen digital safety in Sri Lanka [11],[14],[19].

1.2 Research Gap

The literature review highlights several approaches for detecting harmful content, violence, and harassment on social media [21],[23]. Researchers have utilized natural language processing (NLP) for text analysis [2],[4], computer vision for image and video recognition [7],[9], and audio processing for speech or aggressive sound detection [5], [6]. While these studies demonstrate progress in cyberbullying, violence, and harmful content detection, they exhibit significant limitations; recent work shows that multimodal (text, video, audio) methods improve robustness but still face gaps in language coverage and real-time deployment [11], [12], [19], as evidenced by the three proposed systems outlined in Table 1.2.

Features	Text Analy sis	Video Analysis	Audio Analy sis	Multilin gual Support	Real-Time Alerts	Continuous Learning
Research A	✓	✗	✗	English Only	✗	✗
Research B	✓	✓	✗	English Only	✗	✗
Research C	✓	✓	✓	Limited	✗	✗
Proposed System	✓	✓	✓	Sinhala, English	✓	✓

Table 1.1: Comparison between existing systems

Research A focuses solely on text-based analysis using Natural Language Processing (NLP) techniques, such as CNNs and RNNs, to classify abusive or harassing content in English [21], [23]. While effective at identifying toxic or offensive words and phrases, its scope is narrow since it cannot process videos or audio, where much of today’s harmful content is shared [11]. Moreover, it lacks support for regional languages like Sinhala, which significantly reduces its usability in contexts such as Sri Lanka where bilingual communication is common [14]. Another limitation is the absence of real-time alerting, meaning harmful posts can go undetected for long periods, delaying timely intervention by authorities [11].

Research B is centered on video-based detection using computer vision and deep learning models to identify violent or aggressive behaviors in visual content [7],[9]. It can recognize actions such as fighting or threatening gestures, which are often missed in text-only systems [7], [8]. However, it does not analyze accompanying text or audio, limiting its ability to fully understand the context of an incident; as a result, multimodal approaches are generally preferred for robustness [11], [12]. Like System A, it also lacks Sinhala support and real-time alerting, which reduces its effectiveness in fast-moving, multilingual environments like Sri Lanka’s social media space [11], [14].

Research C combines multiple modalities text, video, and audio into a unified detection framework. This integration improves accuracy by analyzing different forms of content together [11], [12], [19]. While more comprehensive than Systems A and B, many such models are restricted to English-language contexts and do not continuously adapt to new slang, local dialects, or evolving patterns of harmful behavior [21], [23]. Furthermore, they are generally not designed for real-time alerting, which limits their ability to support rapid responses to emerging threats [11], [19].

Proposed System, in contrast, builds on the strengths of all three by enabling multimodal detection across text, video, and audio, while also supporting both Sinhala and English [11], [14], [19]. It uses advanced models like YOLOv8 for video, speech-to-text for audio, and transformers for text analysis [9], [5], [2]. Importantly, the system operates in real time, automatically generating alerts to authorities when harmful content is detected, enabling faster, localized, and more effective intervention compared to existing approaches [11], [19].

1.2 Research Problem

A key scientific issue in modern society is the detection of harmful and abusive content on social media platforms using advanced artificial intelligence techniques. Conventional approaches, which rely on manual monitoring by individuals or small teams, are highly labor-intensive, time-consuming, and prone to human bias. In contexts such as Sri Lanka, where bilingual communication in Sinhala and English is common, these traditional monitoring techniques often fail to capture the complexity and scale of harmful content spreading across platforms. This creates an urgent need for automated and intelligent solutions capable of handling large volumes of data in real time [1], [21], [22].

The cornerstone of harmful content detection has traditionally been text-based Natural Language Processing (NLP) methods. These models are useful for identifying abusive or harassing language in English but are inherently limited in scope. They struggle with regional languages such as Sinhala and cannot capture harmful behavior shared through videos, images, or audio clips. Moreover, the lack of real-time alerting in such systems reduces their effectiveness for timely intervention, allowing harmful content to spread widely before authorities can take action. Incorrect or delayed detection in this context can escalate risks, including misinformation-driven unrest, cyberbullying, or even physical crimes [21].

Artificial intelligence and multimodal deep learning models, on the other hand, provide a more comprehensive solution. By combining NLP for text, speech-to-text models for audio, and vision

models such as YOLOv8 for video and image analysis, these systems can identify subtle cues and patterns across multiple content types. This enables the detection of threats or abusive content even when they are embedded in different modalities, which often escape text-only systems. Furthermore, early detection and real-time alerting are crucial for enabling rapid responses by law enforcement, thereby preventing crimes and protecting public safety [2], [5], [7], [9], [11], [19].

The dynamic changes in social media usage worldwide further highlight the importance of this research issue. With billions of posts being shared daily, including in multilingual and multimedia formats, effective monitoring techniques are required to ensure public safety and trust in digital platforms. The growing role of AI in governance and security also emphasizes the timeliness of developing such solutions. Therefore, the goal of this research is to advance scientific understanding of multimodal AI approaches for harmful content detection in social media, especially within bilingual contexts like Sri Lanka. The outcomes of this study could pave the way for automated systems that enable real-time monitoring and intervention, ultimately strengthening crime prevention and enhancing public safety [1], [14], [21], [22].

2. OBJECTIVES

2.1 Main Objectives

The Main Objective is To develop a real-time, bilingual (Sinhala and English), multimodal AI system capable of detecting harmful content including harassment, killings, elder abuse, child abuse, and violent acts across text, video, and voice on social media, and to provide instant alerts to authorities for timely intervention

2.2 Specific Objectives

1. Data Collection and Preprocessing

- Gather social media posts containing text, video, and audio content in Sinhala and English
- Clean, annotate, and preprocess data to prepare it for training AI models

2. Text Analysis

- Develop NLP models to detect abusive, harassing, or violent language in bilingual text.

3. Video Analysis

- Implement computer vision models (e.g., YOLOv8, Video Swin Transformer) to detect violent or abusive actions in videos.

4. Audio Analysis

- Extract audio from social media videos and convert speech to text using models like Whisper or Wav2Vec2.
- Analyze audio transcripts and sentiment cues to identify harmful content.

5. Multimodal Fusion

- Combine predictions from text, video, and audio analysis into a unified system to enhance detection accuracy.

6. Real-Time Detection and Alerts

- Design the system for low-latency, real-time monitoring.
- Automatically notify moderators or law enforcement when harmful content is detected.

7. Continuous Learning

- Retrain models using newly flagged content to adapt to emerging trends, slang, and evolving abusive behaviors.

3. METHODOLOGY

The proposed system uses AI-driven multimodal analysis integrated with a React Native mobile application to automatically detect harmful content from social media platforms. The methodology involves several key stages

1. Requirement Analysis

- Identify target harmful content categories: harassment, killings/murder, elder abuse, child abuse, and violent behavior.
- Define requirements for real-time monitoring, bilingual text analysis (Sinhala + English), and multimodal support (text, video, audio).
- Ensure compliance with ethical, privacy, and legal constraints regarding social media data collection.

2. Data Collection & Preprocessing

- Data Sources: Use official social media APIs (e.g., Twitter/X API, Facebook Graph API, YouTube API) or web scraping tools where APIs are limited.
- Data Types: Collect posts, captions, comments, shared videos, and audio clips.
- Preprocessing:
 - Text: Tokenization, stopword removal, translation (Sinhala/English).
 - Audio: Convert speech in videos to text using Automatic Speech Recognition (ASR).
 - Video: Extract frames for object/activity recognition.

3. AI Model Development

- Text Analysis (NLP): Fine-tune models (mBERT/XLM-R) for hate speech, abusive, and violent text detection in Sinhala and English.
- Audio Analysis: Use ASR (Wav2Vec2) + emotion detection models to identify abusive tone or threatening speech.
- Video Analysis: Use YOLOv8 / ResNet + LSTM to detect violent/abusive actions (e.g., fighting, weapon use).
- Multimodal Fusion: Combine text, audio, and video analysis into a single classification system for improved accuracy.

4. System Architecture Design

- Backend Services
 - AI models hosted on a cloud server (e.g., AWS, Azure, GCP).
 - A crawler module continuously fetches social media data.
 - A stream processor analyzes posts in near real time.
 - A database stores flagged harmful content with metadata.
- Frontend (React Native App)
 - Provides an interface for law enforcement to view flagged harmful posts.
 - Includes filters by category (e.g., harassment, child abuse).
 - Displays alerts and notifications when new harmful content is detected.
- Alert Mechanism
 - Push notifications to police officers when serious harmful content is identified.

5. Implementation

- Implement data-fetching pipeline using APIs & web crawlers.
- Train and deploy AI models on a backend server.

6. Model Training & Evaluation

- Split datasets into training (70%), validation (15%), testing (15%).
- Evaluate models using accuracy, precision, recall, F1-score.
- Compare unimodal vs multimodal performance.
- Assess system's real-time performance: response time, throughput, and scalability.

7. Deployment & Testing

- Deploy backend services on cloud infrastructure.
- Connect real-time crawler to APIs for continuous monitoring.
- Test React Native app with police officers or simulated law enforcement users.
- Conduct performance testing for real-time content scanning.

8. Continuous Learning & Improvement

- Feedback from police officers used to retrain AI models (active learning).
- Improve Sinhala-English NLP models with domain-specific datasets.
- Optimize for scalability to handle millions of posts daily.

Summary of technologies, techniques, architectures, and algorithms used for the classification of insect infestation in button mushrooms shown in the table below.

Technologies	Python, TensorFlow, PyTorch, React Native, Node.js, Express.js, MongoDB, Flask/FastAPI, OpenCV, HuggingFace Transformers, Firebase, VS Code, Docker
Techniques	Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Speech Recognition, Sentiment Analysis, Multimodal Fusion, Data Augmentation
Algorithms	CNN, RNN, LSTM, BiLSTM, YOLOv8, BERT, RoBERTa, Wav2Vec2.0, FastText
Architectures	VGG-16, VGG-19, ResNet-50, Inception v4, YOLOv5, YOLOv8, Transformer-based Models (BERT, RoBERTa), 3D-CNN (for video), Hybrid Fusion Architecture

Table 3.1: Comparison between existing systems

3.1 System Architecture

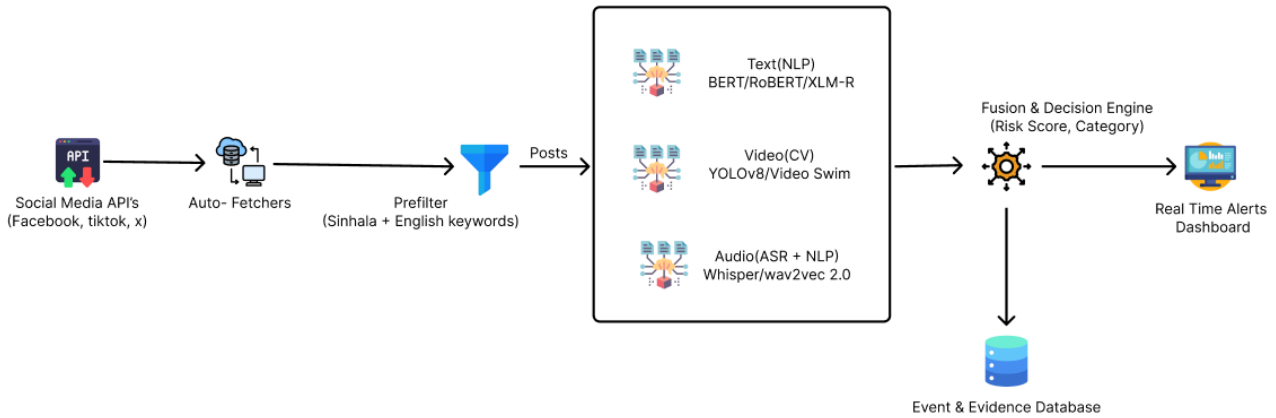


Figure 3.1: System Architecture Diagram

Figure 3.1 presents the end-to-end, real-time pipeline for automatically detecting harmful social-media content without user uploads: Auto-Fetchers continuously watch public platform APIs (Facebook, YouTube, TikTok, X) and pass new items through a Sinhala- and English-based Prefilter, after which each post is analyzed across three parallel modalities—Text (NLP: BERT/RoBERTa/XLM-R) for captions/comments and ASR transcripts, Video (CV: YOLOv8/Video Swin) for violent actions or weapons, and Audio (ASR+NLP: Whisper/wav2vec 2.0) for abusive or threatening speech. Their scores are combined in a Fusion & Decision Engine to produce a unified risk score and category, triggering real-time alerts to the dashboard for police/moderators while archiving frames, transcripts, metrics, and metadata in the Event & Evidence Database for audit, case management, and continuous model improvement.

3.2 Software solution

The proposed system is a real-time multimodal AI-based application that automatically fetches and analyzes social media posts in Sinhala and English to detect harmful categories such as harassment, killings, elder abuse, child abuse, and violent activities. The system consists of three main modules

1. **Data Fetching Layer** – Connects to public social media APIs (e.g., Facebook, YouTube, TikTok, Twitter/X) and continuously fetches posts, comments, and videos based on keyword/hashtag rules and metadata filters.
2. **Multimodal Analysis Layer** – Processes the content using:
 - **Text Analysis:** Natural Language Processing (BERT/XLM-R) for Sinhala and English harmful language detection.
 - **Video Analysis:** Computer Vision models (YOLOv8, Video Swin Transformer) for detecting violent scenes/actions.
 - **Audio Analysis:** Speech-to-Text (Whisper/Wav2Vec2) + NLP classifier, plus sound event detection (screams, gunshots).
3. **Fusion & Alert Layer** – Combines results from all three modalities, assigns a risk score, and sends real-time alerts to law enforcement dashboards, moderation panels, or other stakeholders. The software is designed as a cross-platform web-based system, where authorities can log in to view alerts, evidence, and reports. The backend will be implemented using Python (FastAPI/Flask) with TensorFlow/PyTorch for AI models, and the frontend dashboard will be developed using React/React Native for usability across devices.

- **Requirement gathering**

The requirement gathering process involved the following steps:

- **Survey of Social Media Users**

Conducted to understand how often harmful content is seen and the types most encountered (Figures 1.1 – 1.6). Results showed strong public support for AI-based automated monitoring.
- **Consultation with Law Enforcement**

Discussions with police officials highlighted the delays in manual monitoring of social media posts and the need for an automated early-warning system.
- **Technical Review of Social Media APIs**

Feasibility of connecting to APIs (Facebook Graph API, YouTube Data API, Twitter/X Stream API) was analyzed to determine possible data sources.

- **Research Literature Review**

Examined existing AI-based harmful content detection systems globally and identified the gap for a Sinhala-English, multimodal, real-time solution.

The outcome confirmed the need for a bilingual multimodal AI pipeline that integrates text, video, and audio detection with automatic alerting.

- **Feasibility Study (Planning)**

- **Economic Feasibility**

The system requires initial cloud GPU infrastructure (AWS, GCP, or AZURE) for real-time inference. Compared to manual monitoring, the cost of automation is significantly lower in the long term since fewer human moderators are required.

- **Technical Feasibility**

The technologies required APIs for data fetching, AI frameworks (TensorFlow, PyTorch), NLP models (XLM-R, BERT), CV models (YOLO, Swin Transformer), and ASR models (Whisper) are already available and proven in research. The system will integrate these into a scalable backend using FastAPI + ONNX Runtime/TensorRT for low-latency inference.

- **Operational Feasibility**

Law enforcement and moderators can easily use the web dashboard for reviewing flagged posts. The system includes severity scores, evidence snapshots, and audit trails, ensuring transparency and accountability.

- **Implementation (Development)**

Below are the sub-functionalities that will be implemented during the development phase

- Implementation of real-time social media fetching module to collect posts, comments, and videos from platforms such as Facebook, YouTube, TikTok, and Twitter/X.
- Development of a keyword-based pre-filtering system in Sinhala and English to reduce irrelevant data before AI analysis.
- Implementation of text classification module using Natural Language Processing (BERT/XLM-R) to detect harassment, hate speech, and abusive language.
- Implementation of video analysis module using Computer Vision models (YOLOv8, Video Swin Transformer) to identify violent actions, weapons, and abusive behavior.
- Implementation of audio analysis module using speech-to-text (Whisper/Wav2Vec2) and NLP models to detect abusive language, screams, and threats from voice content.
- Integration of multimodal fusion engine to combine text, video, and audio outputs into a unified decision with severity levels.

- Development of alerting system and dashboard to notify law enforcement officers and moderators in real time with flagged content, evidence, and severity levels

- **Testing**

- Verifying that the system has all the requirements as expected.
- Testing for errors and bugs.

The testing of the application will be conducted in various stages, including unit testing, integration testing, system testing, and acceptance testing, with the latter being carried out by the user.

3.3 Future scope

In the future, the system can be improved with more features and wider use. The AI models can be updated regularly with new data to keep up with changing slang, language, and harmful behaviors. The system can be extended to work in more languages and scaled to monitor larger amounts of social media content in real time. A mobile version can also be developed so that police officers and moderators can receive alerts instantly on their devices. Cloud deployment will allow the system to handle national level data and provide faster performance.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The system should be able to fetch posts, comments, and videos from social media platforms in real time.
- The system should analyze text content (Sinhala and English) to detect harassment, hate speech, and abuse.
- The system should analyze video content to detect violent actions or abusive behavior.
- The system should analyze audio content to detect harmful speech, threats, or violent sounds.
- The system should combine results from text, video, and audio to decide if the post is harmful.
- The system should send real-time alerts to law enforcement dashboards or moderators when harmful content is detected.
- The system should store flagged posts with evidence (text, frames, transcripts) for future review.
- The system should allow law enforcement or moderators to review, approve, or reject flagged cases.

4.2 Non-functional Requirements

- User-friendliness – The system should have a simple and easy-to-use dashboard for police officers and moderators, with clear alerts and case details.
- Reliability – The system should run without crashes or errors and protect all confidential information securely.
- Performance – The system should process social media posts quickly and provide real-time alerts with low latency.
- Scalability – The system should be able to handle large volumes of posts and scale up as the number of users or data increases.
- Availability – The system should be available 24/7 so that harmful content is detected at all times.
- Security – The system should ensure that all data, especially sensitive evidence, is stored and transmitted securely with access only for authorized users.

4.3 System requirements

Software Requirements

1. Python – For backend development and AI model implementation.
2. TensorFlow / PyTorch – For training and running multimodal AI models.
3. FastAPI / Flask – To build the backend API for real-time processing.
4. React / React Native – For building the web dashboard and mobile access.
5. Database (MySQL / MongoDB) – For storing flagged posts, evidence, and user data.
6. Cloud Services (AWS / GCP / Azure) – For deployment, GPU processing, and scalability.

4.4 User Requirements

1. Police Officers / Law Enforcement

- Should be able to log in securely to the system.
- Should receive real-time alerts when harmful content is detected.
- Should view flagged posts with evidence (text, video frames, transcripts).
- Should mark cases as reviewed, investigated, or resolved.

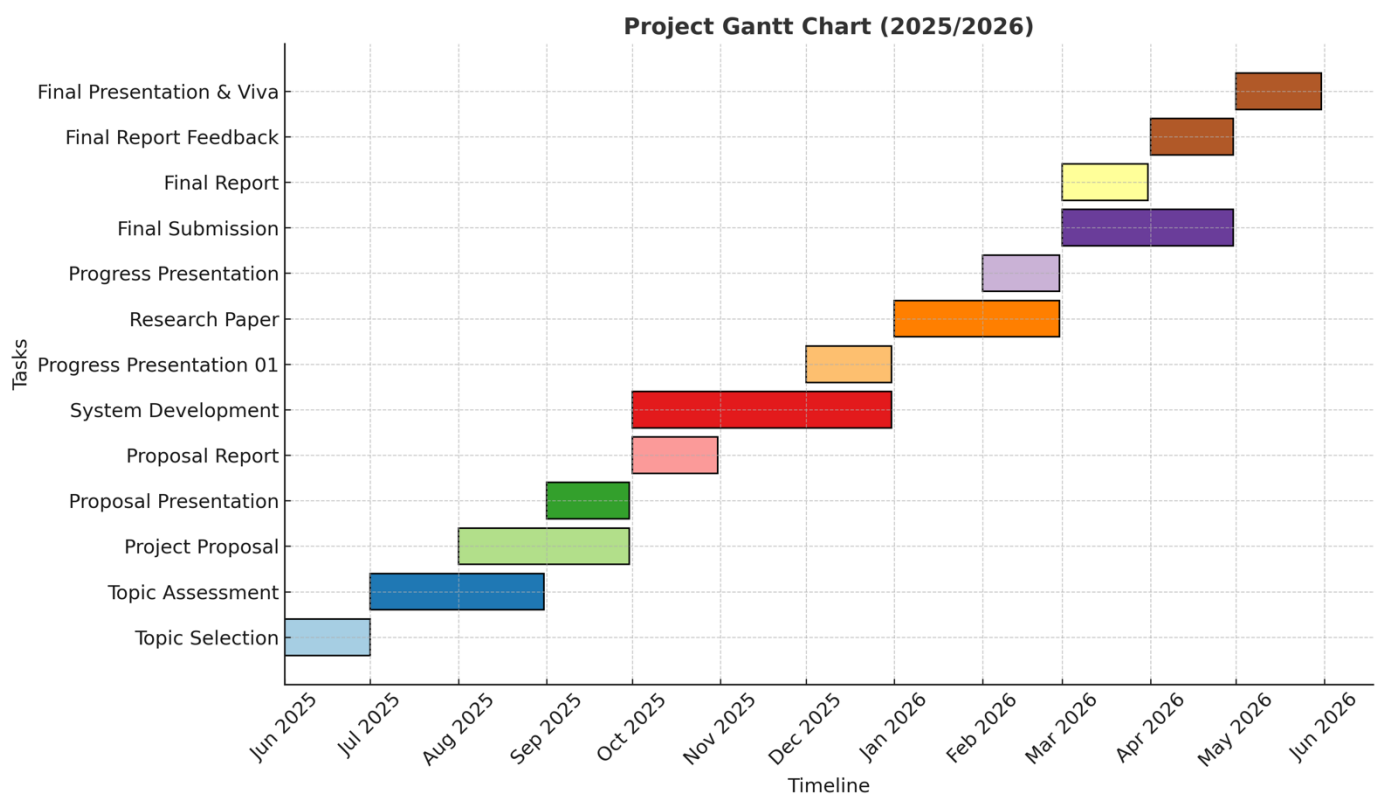
2. Moderators / Analysts

- Should be able to monitor all flagged posts and verify system detections.
- Should approve or reject harmful content alerts suggested by the AI.
- Should provide feedback to improve system accuracy.

3. System Administrators

- Should manage users and access rights.
- Should maintain the system, update models, and monitor system performance.
- Should ensure data security and handle backups.

5. GANTT CHART



6. BUDGET AND BUGET JUSTIFICATION

Below table 6.1 shows the overall budget for the overall proposed system.

Expenses	
Expense	Approx. Cost (LKR)
Data Collection	20,000.00
Cloud/GPU (Colab Pro)	15,000.00
Deployment/Hosting	10,000.00
Total	45,000.00

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